MIKAEL CHRISTODOULIDES

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Postgraduate honours student with a keen interest in energy materials and pure mathematics. Eager to pursue further studies focusing on innovations in sustainable energy, and the use of algebra to drive inventions in materials engineering.

Education

2026 **Bachelor of Science Honours in Materials Science, UCT**

Currently in-progress.

2023–2025 **Bachelor of Science in Mathematics, Chemistry, Computer Science, UCT**

Received Tier 1 Science Faculty Entrance Scholarship (2023), as well as Faculty Scholarships all subsequent years (2024, 2025). Placed on the Dean's Merit List for academic excellence across all years (2023, 2024, 2025). Additionally, received two class medals for attaining the highest mark in courses MAM1019H, CSC2005Z.

2018–2022 **Westerford High School**

Received academic honours for all years (2020, 2021, 2022). Placed in the Top 20 matric students with a 94% average, and received the Student Award for Music (2022).

Research Experience

2026 **Using Symmetry Groups to Model Damage in Crystalline Energy Metals (Working Title)**

M. Christodoulides, T. Becker (Supervisor)

Currently in-progress as a year-long honours thesis.

2026 **Experimental investigation on the response of sandwich panels with 3D printed biopolymer gyroid structure cores and asymmetric glass fibre face sheets under blast loading**

J. A. Dicks, S. Schlesinger, M. Christodoulides, S. Gabriel and L. Lthuli

Co-authored the above paper, completed in conjunction with and to serve as a supplementary resource for, the paper below.

2026 **3D printed photopolymer interpenetrating gyroid lattice structures and application in sustainable sandwich composites**

J. A. Dicks, S. Gabriel, M. Christodoulides and L. Lthuli

Co-authored the above paper, for submission to ICILSM 2026, the 5th International Conference on Impact Loading of Structures and Materials.

2025 **An Analysis of Temporal Digraphs Modeled From Hurricane Paths as Inspiration for Studying the Effect of Edge Weights on a Graph's Structure**

M. Christodoulides, D. Erwin (Supervisor)

Written for MAM3006Z, an optional mathematics research course completed during third year of study.

2024

An Analysis of Randomness in Computer Science in the context of a Truly Random Number Generator

M. Christodoulides, P. Marais (Supervisor)

Written for CSC2005Z, an optional computer science research course completed during second year of study.

Work Experience

Jan 2026

Research Assistant, UCT

Worked as a research assistant in the Centre for Materials Engineering for a month during January/February 2026. Assisted with manufacturing and research in sustainable composite materials and the use of 3D-printed cores for blast-resistant panels.

Dec 2025

Volunteer, Hostel20 Bayonne

Volunteered at a hostel in Bayonne, France, for a month during December 2025. Improved French proficiency and gained experience working in hospitality environments.

2024–2026

Mathematics Tutor, UCT

Worked as a departmental tutor teaching 1st year students in the science and engineering faculties. Received overwhelmingly positive feedback from students' course evaluations about approachability and clarity of explanation throughout time of tutoring. Currently completing third year of work.

Skills

- **Languages:** English (fluent), French (intermediate)
- **Software:** LaTeX, Java, C#, Python, HTML

References

- Thornsten Becker, Engineering Professor, UCT
- Anneliese Schuarte, Mathematics Assoc. Professor, UCT